

Clarence Cheong

Systems Architect | AI Solutions Engineer | Computer Vision

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Professional Summary

Systems-driven Solutions Architect with over a decade of multi-disciplinary experience spanning Computer Vision (CV), Enterprise Software, Cloud Infrastructure, and Legacy Mainframe integration. Specialized in designing modular, extensible architectures that decouple core business flows from underlying service providers — treating each component as a black-box service with predefined input/output contracts, so new providers and methods can be plugged in without refactoring the main pipeline. Proven track record of bridging deep-tech research and production-grade, distributed deployments — from custom LLM RAG frameworks on GCP to large-scale Smart City video analytics engines — while orchestrating containerized environments and edge-AI runtimes.

Experience

Almex System Technology Asia (Almex STA)

Senior AI Solutions Engineer

Remote, MY.

May 2019 – present

Python, Docker, Bash, OpenCV, FastAPI, OpenVINO, GCP, Ansible, TensorFlow, Flask, MongoDB, PyQt, Linux, Pydantic.

- Worked remotely with a team in Tokyo, Japan.
- Architected Enterprise RAG Infrastructure: Designed the end-to-end architecture of a GCP-hosted RAG chatbot ecosystem around a modular, provider-agnostic core. Defined stable input/output contracts — enforced with Pydantic BaseModel schemas — that treat each LLM provider, vector store, and retrieval/generation stage as a pluggable black-box service, decoupling the main pipeline flow from any single vendor so providers and methods can be swapped or extended without refactoring the core flow. Delivered secure, scalable API endpoints for dynamic knowledge-base updates and an automated evaluation framework auditing relevance, faithfulness, and hallucination mitigation.
- Designed CV Pipelines from POC to Production: Formulated technical proposals and system designs for mission-critical CV products, including an Invoice OCR document-classification engine and a Person Re-Identification system demonstrated at international trade fairs. Structured each pipeline with standardized service boundaries and I/O contracts so interchangeable processing backends could be swapped while the orchestration flow remained unchanged.
- Optimized Edge AI Deployment & System Integration: Co-authored high-performance API integrations for FLIR AX8 Thermal Cameras featuring real-time face detection and blackbody-calibrated thermal corrections. Optimized deep learning models utilizing Intel's OpenVINO toolkit for deployment at the edge.
- Automated Cross-Workstation Orchestration: Overhauled an internal image-labeling application by implementing a MongoDB data layer and integrating automated text-detection algorithms. Streamlined installation friction across multiple distributed workstations via Ansible configuration management.
- Engineered Complex ML Data Synthesis Systems: Built custom ML data tools, including a synthetic Japanese character OCR training system utilizing font glyph checking and canvas-overflow validation, an enhanced SynthText background-generation engine manipulating depth and segmentation maps, and an FCOS-styled tfrecord translation pipeline.

Multimedia University

Computer Vision Research Scholar

Cyberjaya, MY

Oct 2016 – Apr 2019

C++, OpenCV, Python, JavaScript, Linux.

- Engineered Video Analytics Frameworks: Directed semantic extraction and data preprocessing workflows for project SHERLOCK, a large-scale, multi-camera video analytics initiative optimized for smart city long-term surveillance.
- IP Creation & Research Leadership: Successfully obtained intellectual property copyright for the developed core vehicle semantic retrieval engine. Published 3 peer-reviewed IEEE/Springer international conference papers and presented findings to industry stakeholders.

Freelance & International Research Consultant

Freelance & Intern

Malaysia & Taiwan

June 2018 – Feb 2019

AWS (EC2/S3), Docker, Raspberry Pi, Keras, TensorFlow, OpenCV, Python.

- Retail Store Analytics Solution (Freelance): Designed and deployed an edge-to-cloud customer analytics pipeline using Raspberry Pi nodes and custom deep learning models. Streamlined gender and behavioral inference processing on the edge before structured JSON data payloads were shipped to AWS cloud buckets for client business-intelligence consumption.
- Vehicle Trajectory Analytics (Research Intern, National Chung Cheng University, TW): Researched and prototyped deep CNN architectures to handle automated vehicle trajectory classification under the supervision of Professor Wen-Nung Lie.

Hewlett Packard Enterprise (HPE)

Application Management & Service Delivery Engineer

Cyberjaya, MY

Sept 2012 – Sept 2016

Mainframe z/OS, COBOL, JCL, CA-7, Bash, SQL, Visual Basic.

- High-Availability Production Systems Support: Provided Tier-2 24/7 technical incident and crisis management for CIMB Bank's core payment channels (including IBG, Autopay, Remittance, and Direct Debit), ensuring zero-downtime operations for critical financial traffic.
- Cross-Platform Systems Engineering: Interfaced directly with client stakeholders and IT operations teams to implement data extraction applications for the enterprise-wide 'CIMB 1 Platform' banking migration project.

Multimedia University

Research Assistant & Web Systems Developer

Cyberjaya, MY

Aug 2010 – Jan 2012

PHP, SQL, HTML/CSS, WAMP Server.

- Designed, implemented, and managed a centralized Content Management System (CMS) utilized by university staff to audit and track R&D progress. Successfully launched the software into active production, sustaining a 5-year operational lifecycle until retirement.

Core Competencies

System & Solution Architecture: Modular & Extensible System Design, Abstraction & Adapter Layers, Contract-First Service Design (Black-Box I/O, Pydantic Schema Validation), Provider-Agnostic Integrations, Decoupling Business Flow from Service Providers, Microservices & Distributed Systems

Architectures: End-to-End AI/ML Pipelines, RAG Frameworks, Distributed Systems, Microservices, Edge AI

Computer Vision & Deep Learning: Object Detection/Tracking, Person Re-Identification, OCR (Synthetic Data Gen), OpenVINO, TensorFlow, Keras

DevOps & Cloud Infrastructure: Docker, Ansible, GCP, AWS, CI/CD, Shell Scripting, Linux System Admin

Enterprise Integration: Enterprise APIs, Relational & NoSQL Databases (SQL, MongoDB), Mainframe (COBOL, JCL, z/OS), ITIL Incident & Crisis Management

Full-Stack & Automation: Python, FastAPI, C++, Flask, JavaScript, PHP, Visual Basic, Automations, PyQt GUI Development

Education

MSc	Multimedia University , Information Technology	Cyberjaya, MY
	• Thesis: Extraction and Retrieval of Object Semantics for Long-Term Car Park Surveillance Videos • Published 3 papers, and won multiple awards.	Oct 2016 – May 2019
BEng	Multimedia University , Electronics Majoring in Multimedia	Cyberjaya, MY
	• Thesis: Multiple Speaker Voice Recognition	June 2008 – June 2012
Dip	Tengku Abdul Rahman College (TARC) , Mechatronics	Kuala Lumpur, MY
		June 2006 – May 2008

Publications

Efficient Semantic-Based Vehicle Retrieval In Long-Term Car Park Videos	July 2019
<i>Clarence Cheong</i> , Ryan Woei-Sheng Lim, John See, Lai-Kuan Wong, Ian K.T. Tan 10.1109/ICMEW.2019.00-97 (ICMEW 2019)	
Vehicle Semantics Extraction and Retrieval for Long-term Carpark Video Surveillance	Feb 2018
<i>Clarence Cheong</i> , Ryan Woei-Sheng Lim, John See, Lai-Kuan Wong, Ian K.T. Tan, Azrin Aris 10.1007/978-3-319-73600-6_27 (MMM 2018)	
On vehicle state tracking for long-term carpark video surveillance	Sept 2017
Ryan Woei-Sheng Lim, <i>Clarence Cheong</i> , John See, Ian K.T. Tan, Lai-Kuan Wong, Huai-Qian Khor 10.1109/ICSIPA.2017.8120638 (ICSIPA 2017)	

Selected Honors

- International Invention, Innovation & Technology Exhibition (ITEX) 2018: [ICT & Multimedia - University & Research Institute Category] (Silver Medalist)
- Malaysia Technology Expo (MTE) 2018: [Invention & Innovation Category] (Bronze Medalist)
- A.I. Hackathon for Good (2nd Runner Up)

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1. Core Retrieval Engine for Smart City Analytics